

Namaste React

Episode 01 - Inception

What is a CDN ?

- A Content Delivery Network or Content Distribution Network (CDN) is a geographically distributed network of proxy servers and their data centers.
- The goal is to provide high availability and performance by distributing the service spatially relative to end users.
- CDNs have grown to serve a large portion of the internet content today, including web objects (text, graphics & scripts), downloadable objects (media files, software, documents), applications (e-commerce, portals), live streaming media, on-demand streaming media, and social media sites.
- CDNs are a layer in the internet ecosystem. Content owners such as media companies and ecommerce vendors pay CDN operators to pay their content to their end users.
- We can add React into our project by injecting CDN links in it (in an .html file).

Why do we use CDN ?

- Improved scalability and connectivity.
- In addition to facilitating end-users with faster load times, which translates into greater user -experience, a content delivery

network also rewards web publishers with increased traffic, higher page views, etc.

- Decreased bandwidth consumption.
- Lower latency.
- Latency is the lag between request and response.
- Effective traffic spike management.
- Enhanced cyber security.
- ACDN employs automation and data analytics tools that help identify firewall issues, Man in the middle threat, Distributed Denial of Service attacks.

What is crossorigin and why do we use it in React CDN ?

- CORS or cross-origin resource sharing is a mechanism that allows memory resources (e.g., fonts, JavaScript, etc) on a webpage to be requested from another domain outside the domain from which the resource originated.

React.development.js

- This file is the core of React.
- This file contains the whole code of React which is written in JavaScript.

React-dom.development.js

- Using this file, React interacts with the browser DOM.

First program in React

```
import React from "react";
import ReactDOM from "react-dom";

const Heading = React.createElement(
  "h1",
  { id: "heading" },
  "Hello world from React app!"
);

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<Heading />);
```

- The costliest operation for a browser is when the browser needs to manipulate the DOM.
- Any React element is nothing but a JavaScript object.
- This object contains a key known as *props* which stores the children and other attributes of the React element in a key-value pair.
- The `render()` function is responsible for taking the JavaScript object (React element) as an argument, converting it into an HTML tag and putting it in the DOM.